WorldPop

Workshop on Advanced Sampling Methodologies

Welcome and Introduction

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Welcome!

Team

Dr. Sarchil Quader

- Senior Enterprise Fellow at WorldPop, University of Southampton.
- Develop research to improve population mapping and agricultural monitoring in low- and middle-income countries using big data, such as remote sensing.

Dr. Gianluca Boo

- Senior Research Fellow at WorldPop, University of Southampton.
- Develop research to improve population mapping in low- and middle-income countries using statistical modelling, particularly Bayesian hierarchical models.

Workshop

Objectives

- Strengthen **geospatial and sampling** capacity of Pakistan Bureau of Statistics (PBS).
- Introduce AI-assisted sampling to **improve efficiency, representativeness, and inclusivity**.
- Align **methods with Government priorities** and international standards for comparability.
- Enhance **policy relevance of PDHS data** and enable cross-country analysis.

Workshop

Topics

- Foundations of **survey sampling** theory and practice.
- **Geospatial data** processing and updating census/sampling frames.
- Use of **Earth Observation (EO)** data for spatial stratification.
- Application of **AI/ML methods** (clustering, prediction) in sample design.
- **Simulation and optimization** of sampling strategies using R.
- Development of R Shiny applications to **evaluate and visualize sampling designs**.
- Hands-on practice with **survey analysis software** and calibration techniques.
- Addressing **cost-effectiveness and nonresponse** adjustments in survey design.

Workshop

Expected Outcomes

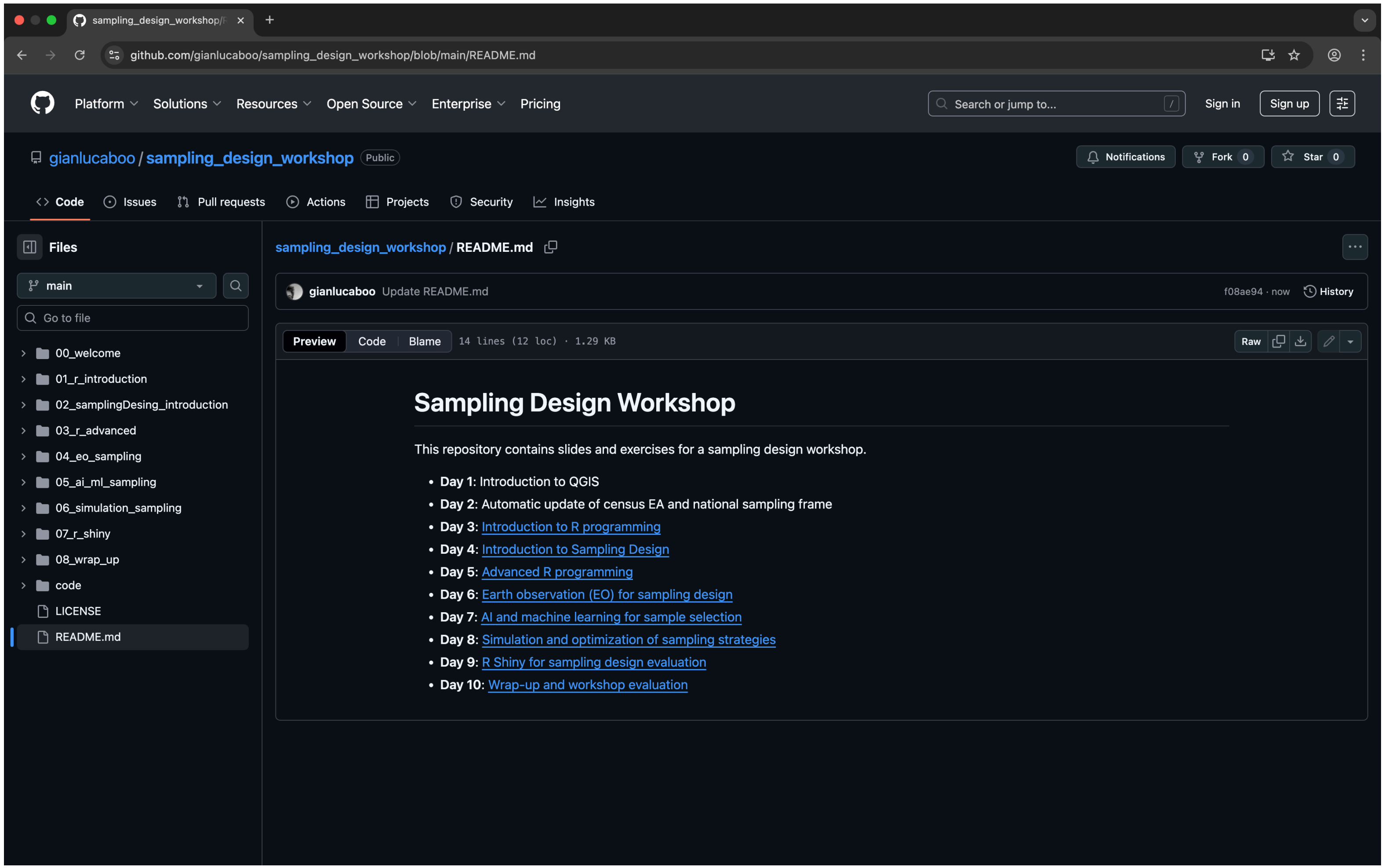
- Enhanced PBS ability to **design robust, adaptive, and cost-effective samples**.
- Practical **skills in R, GIS, EO data, and AI/ML tools**.
- Prototype **interactive tools (Shiny apps) for survey design evaluation**.
- Improved readiness to implement the upcoming PDHS with **modern, AI-driven approaches**.

Agenda

- **Day 1**: Introduction to QGIS
- **Day 2**: Automatic update of census EA and national sampling frame
- **Day 3**: Introduction to R programming
- **Day 4**: Introduction to Sampling Design
- **Day 5**: Advanced R programming
- **Day 6**: Earth observation (EO) for sampling design
- **Day 7**: AI and machine learning for sample selection
- **Day 8**: Simulation and optimization of sampling strategies
- **Day 9**: R Shiny for sampling design evaluation
- **Day 10**: Wrap-up and workshop evaluation

Course materials

All the course materials are available on a dedicated **GitHub repository**.



Let's get started!